

The dynamical hierarchy is an invariant object S of a dynamical system (X, T, φ_t) — each column spans exactly the properties established for it; a gap is a genuine no (or beyond the notes)



Foundations

§1

Dynamical system (X, T, φ_t)
phase space X ; time set $T \subseteq \mathbb{R}$ or $\mathbb{Z}$; evolution operators $\varphi_t \colon X \rightarrow X$ with **(D1)** $\varphi_0 = \operatorname{Id}$ and **(D2)** $\varphi_t \circ \varphi_s = \varphi_{t+s}$

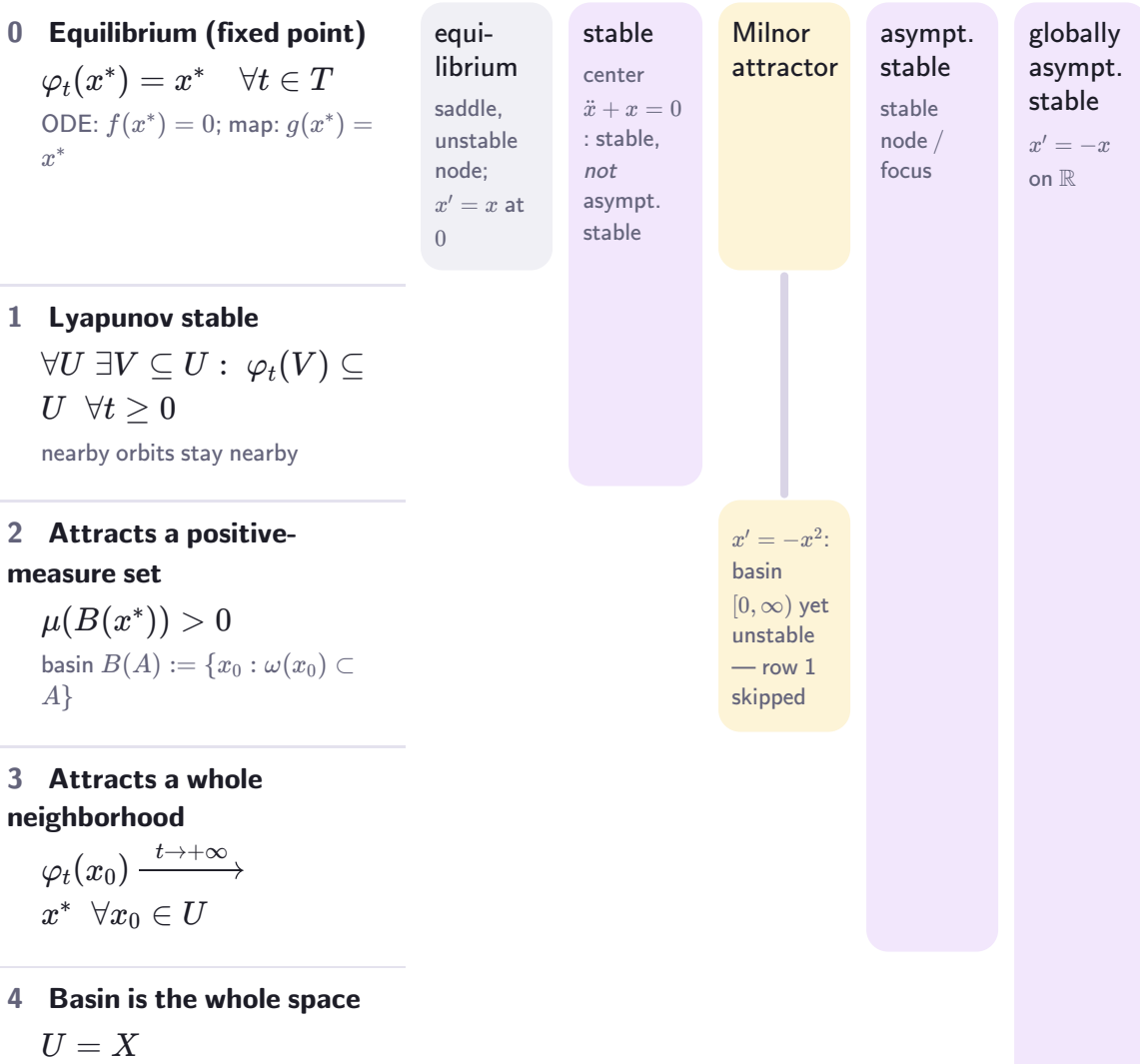
Kinds of time
discrete: $T \cong \mathbb{N}_0$, $\mathbb{Z}$ — iterated maps g^k ; **continuous:** T an interval — flows; **invertible:** T a group, $\varphi_{-t} = \varphi_t^{-1}$.
logistic map $\operatorname{pr}(1 - x)$: inverse is two-valued $\Rightarrow$ not invertible

Orbits & invariant sets
 $\operatorname{Or}(x_0) = \{\varphi_t(x_0) \colon t \in T\}$; periodic: $\varphi_{t_0}x_0 = \varphi_0(x_0)$; the **phase portrait** is the decomposition of X into orbits.
 S invariant: $\varphi_t(x_0) \in S$ for all t , all $x_0 \in S$ (**positively/negatively:** $t > 0 / t < 0$).
nullclines of $x' = f(x)$: $\{x \colon f_A(x) = 0\}$ — organize the phase portrait, van der Pol: $x' = y - \frac{x}{3} + x^3$, $x, y' = -yx$: cubic & vertical line

Existence & uniqueness
 f smooth $\Rightarrow x' = f(x)$, $x(0) = x_0$ has a unique smooth solution on a maximal interval $(-t_*, t_*)$ — via Banach's fixed-point theorem (conjugation on a complete metric space).
maximal $\neq$ global: $x' = x^2$ blows up, $\varphi_t(x_0) = \frac{x_0}{1-tx_0}$ — only a local flow

Stability of equilibria

§2



Stability certificates

§2

Lyapunov function on $U \ni x^*$
• **(D1)** L continuous
• **(D2)** $L(x^*) = 0$, $L(x) > 0$ for $x \neq x^*$
• **(D3)** nonincreasing along orbits: $t_0 < t_1 \Rightarrow L(\varphi_{t_1}(x_0)) \geq L(\varphi_{t_0}(x_0))$ (strict: never =)
differentiable case: $\frac{d}{dt}L(\varphi_t(x)) = \nabla L \cdot f \leq 0$ (Lie derivative)

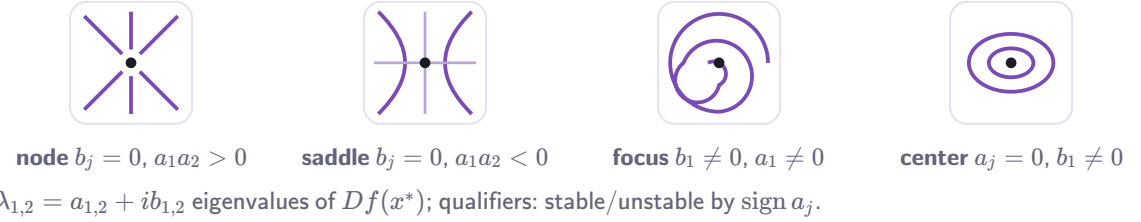
Lyapunov criterion & LaSalle
 $\exists$ Lyapunov function $\Rightarrow$ **stable** (sublevel sets S_c are positively invariant)
 L not constant on any nontrivial orbit $\Rightarrow$ **asymptotically stable** (L is constant on $\omega(x)$, so $\omega(x) = \{x^*\}$)

ω/α -limit sets
 $\omega(x) = \{y \colon \exists t_0 \rightarrow +\infty, \varphi_{t_0}(x) \rightarrow y\}$, $\alpha_t(x) \rightarrow -\infty$

First integral constant of motion
 $\forall g, f = 0$. Hamiltonian $x' = \partial_x H$, $y' = -\partial_x H$ has $H = \frac{1}{2}|y|^2 + V(x)$ as first integral (not a Lyapunov function in general).
homoclinic oscillator: $L = x^2 + y^2$ is both

Linear(zed) stability
 $x' = Ax$: all $\operatorname{Re} \lambda_j < 0 \Rightarrow$ asympt. stable, some $\operatorname{Re} \lambda_j > 0 \Rightarrow$ unstable.
maps $x_{i+1} = Ax_i$: multipliers $|\mu_j| < 1 \Rightarrow$ asympt. stable.
 $\operatorname{Re} \lambda_j = 0$ cases undecided by linearization $\Rightarrow$ center manifold

Worked example: 1D bifurcation analysis exam pattern
 $x' = f(x, p) = x + 2x^2 - x^3 = x(p + 2x - x^2)$
• **steady states:** $x = 0$ and $x_{\pm} = 1 \pm \sqrt{1-p}$ (exists for $p \geq -1$)
• **linear stability:** $\partial_x f = p + 4x - 3x^2$, at $x = 0$: $\lambda = p$ — stable $p < 0$, unstable $p > 0$
• **fold** at $(x, p) = (1, -1)$: $f = 0$, $\partial_x f = 0$, $\partial_{xx} f = -2 \neq 0$, $\partial_p f = 1 \neq 0$ ✓
• **transcritical** at $(0, 0)$: x crosses 0, stability is exchanged
• sketch: branches = 0 and the parabola $p = x^2 - 2x$ in the (p, x) -plane, solid = stable, dashed = unstable



Hyperbolicity & invariant manifolds

§3, §10

flows: x^* hyperbolic $\iff Df(x^*)$ has no eigenvalue with $\operatorname{Re} \lambda = 0$
 maps: x^* hyperbolic $\iff$ all multipliers $|\mu_j| \neq 1$ of $Dg(x^*)$

Manifolds in 60 seconds §3
 d -dim **topological manifold**: locally homeomorphic to $\mathbb{R}^d$ (second-countable, Hausdorff); C^k -**manifold**: atlas with C^k chart transitions $h_\alpha \circ h_\beta^{-1}$. **Tangent space** $T_x M$: velocities $\frac{d}{dt} f(\gamma(t))|_{t=0}$ of curves through x .
every $d \leq 3$ topological manifold is smoothable — false for $d \geq 4$, examples: $\mathbb{R}^4$, $\mathbb{T}^4 = \mathbb{R}^4/2\mathbb{Z}^4$, S^2

Invariant splitting
 $E^s = \{y \colon e^{At} y \rightarrow 0, t \rightarrow +\infty\}$, $E^u = \{y \colon t \rightarrow -\infty\}$; $\mathbb{R}^d = E^s \oplus E^u$ with projections P^s, P^u , $E^c = \{y \colon e^{At} y = y, t \rightarrow +\infty\}$ transverse to the flow, Implicit function theorem $\Rightarrow$ return time $\tau \in C^1$, $\tau(y) = T$, and,
 $P(x) = \varphi_{\tau(x)}(x) \in C^1(S \cap B(y, \delta))$

Stable/unstable manifold theorem flows & diffeos, $f \in C^k$
 $W_{loc}^s(x^*) = \{y \in U \colon \varphi_t(y) \in U \quad \forall t \geq 0, \varphi_t(y) \rightarrow x^*\}$
constructions: C^k manifolds, **tangent** to $E^{s,u}$ at x^* ; globalize by $W^{s,u} = \bigcup_t \varphi_t(W_{loc}^{s,u})$.
constructions: Lyapunov–Perron (variation of constants + contraction) or Hadamard graph transform; maps $W_{loc}^s = \{y \colon y'(t) \rightarrow x^*\}$.
ex. Duffing $\ddot{x} = x - px^3$: saddle at 0, $\dim E^s = \dim E^u = 1$; Arnold cat map on $\mathbb{T}^2$; every point hyperbolic

Hartman–Grobman
 x^* hyperbolic $\Rightarrow \varphi_t$ is **locally topologically conjugate** to the linearization e^{At} , $A = Df(x^*)$.
Maps: $h \colon A = g \circ h, h = \operatorname{Id} + H$ unique via Banach fixed point.
 $\Rightarrow$ near hyperbolic points, the linear picture is the whole local picture (up to homeomorphism — see equivalence ladder)

Non-hyperbolic? — center manifolds

Spectral split
 d_c eigenvalues $\operatorname{Re} \lambda = 0$, d_s with $\operatorname{Re} \lambda < 0$, d_u with $\operatorname{Re} \lambda > 0$; E^c = span of critical eigenvectors.

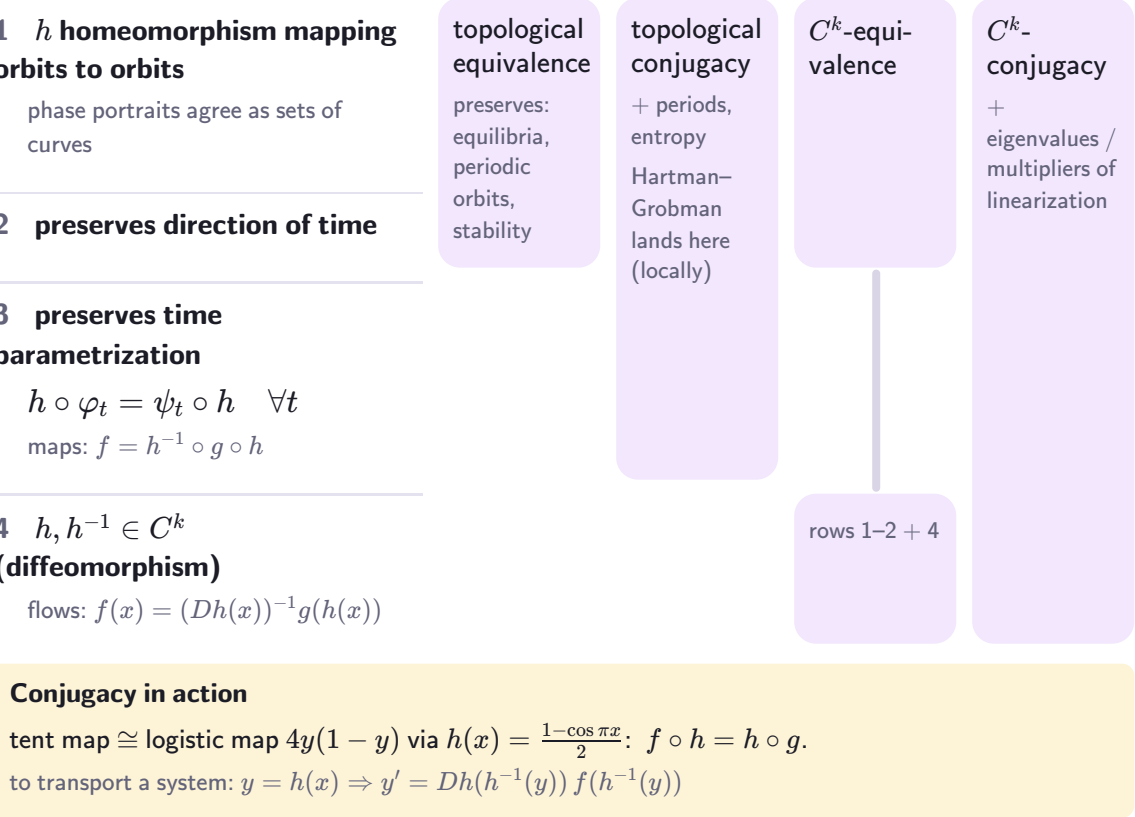
Center manifold theorem $f \in C^k$
 $\exists$ local C^k , d_c -dim invariant manifold $W_{loc}^c(0)$ tangent to E^c ; solutions staying near 0 for $t \rightarrow \pm\infty$ lie on it.
caveats: not unique, possibly not analytic; not C^∞ — e.g. $x' = x^2$, $y' = -y$ has the family $g = pe^{t/\alpha}$ ($x < 0$)

Reduction theorem
With blocks $x' = Bx + F(x, y)$, $y' = Cy + G(x, y)$ ($x \in \mathbb{R}^d$): locally topologically equivalent to
 $x' = Bx$, $y' = Cy + F(x, H(x))$, $y' = Cy$
where $W_{loc}^c = \operatorname{graph} H$, $H = O(x^2)$.

Recipe: computing W^c exam worksheet
• shift equilibrium to 0, split into $B(x, B)$ blocks (diagonalize)
• ansatz $y = h(x) = a_2 x^2 + a_3 x^3 + \dots$
• invariance equation $N(h) = Dh(x)Bx + F(x, h) - Ch - G(x, h) = 0$
• match coefficients; approximation thm: $N(h) = O(|x|^p) \Rightarrow H = h + O(|x|^{p+1})$
ex. $x' = xy$, $y' = -y + px^2$: $a_2 = p$, $W^c = \{y = px^2 + O(x^3)\}$, reduced flow $x' = px^2 + O(x^3)$.

The equivalence ladder

§4



Hyperbolic sets

§7

Hyperbolic set $\varphi \in C^1$ diffeo, λ compact invariant, $\exists C > 0, \lambda \in (0, 1)$ and subspaces $E^s(x)$, $E^u(x) \subset T_x M$:
• **(D1)** invariance: $Dg_t E^{s,u}(x) = E^{s,u}(g(t)x)$
• **(D2)** $|Dg_t v| \leq C\lambda^t |v|$ on E^s , $\lambda \geq 0$
• **(D3)** $|Dg_t v| \leq C\lambda^t |v|$ on E^u
• **(D4)** splitting: $T_x M = E^s(x) \oplus E^u(x)$
Anosov diffeomorphism: $M = \Lambda$, ex. cat map $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ on $\mathbb{T}^2$.
• count orbits systematically
• countably many (saddle) periodic orbits, of every period
• uncountably many aperiodic orbits; a dense orbit
• X is a hyperbolic set $\Rightarrow$ the whole picture **persists** under perturbation
• g_t is chaotic: Lyapunov exponents $\ln \mu > 0 > \ln \lambda$
alternative entrance: Shilnikov saddle-focus loop (zoo) also forces horseshoes

Shadowing
 δ -pseudo-orbit: $d(g(x_i), x_{i+1}) \leq \delta$. Near a hyperbolic A , $\forall \varepsilon \exists \delta$ such that every δ -pseudo-orbit is ε -shadowed by a true orbit.
numerics with rounding errors still trace real orbits

Persistence
 C^1 -small perturbations g of g : nearby compact invariant sets remain hyperbolic, and $\exists$ hyperbolic K with $\psi \circ f|_K = g|_K \circ \psi$, ψ homeo near Id .
Cor: Anosov diffeos are open in C^1 and structurally stable.
the horseshoe's invariant set is hyperbolic $\Rightarrow$ chaos persists — see the road to chaos

Flows $\leftrightarrow$ maps

§5

continuous time φ_t equilibrium $f(x^*) = 0$	discrete time g^k fixed point $g(x^*) = x^*$
asympt. stable $\Rightarrow \operatorname{Re} \lambda_j < 0$	$\mu_j < 1$
periodic orbit Γ, period T	fixed point of Poincaré map P
time-T map φ_T	any diffeo, via suspension flow

Poincaré map at $y \in \Gamma$
section $\Sigma = \{x \colon (x - y) \cdot f(y) = 0\}$ transverse to the flow, Implicit function theorem $\Rightarrow$ return time $\tau \in C^1$, $\tau(y) = T$, and,
 $P(x) = \varphi_{\tau(x)}(x) \in C^1(S \cap B(y, \delta))$

Multipliers & monodromy
multipliers $\mu_1, \dots, \mu_{d-1} =$ eigenvalues of $DP(0)$ — independent of the point on Γ , the section, and coordinates (local Poincaré maps are conjugate).
variational equation $v' = Df(\Gamma(t))v$; monodromy matrix $M(T)$ has eigenvalues $\{\mu_1, \dots, \mu_{d-1}\}$.
all $|\mu_j| < 1 \Rightarrow \Gamma$ asymptotically stable.

Suspension & obstruction
 $g(x) = -\frac{x}{2}$ is not a time- T map of any flow on $\mathbb{R}$ (orientation). But every diffeo g of $\mathbb{R}^d$ is the return map of a flow on the $(d-1)$ -dim mapping torus $M = (\mathbb{R}^d \times [0, T])/\sim$, $(g(x), 0) \sim (x, T)$.

Structural stability

§6

Definition
 $f \in C^k(M)$ is **structurally stable** if $\exists \varepsilon > 0$: every g with $\|f - g\|_C < \varepsilon$ generates a flow topologically equivalent to g^t .
 $\|f\|_C = \sum_{|\alpha| \leq k} \sup_{\mathbb{R}^d} |D^\alpha f|$ on compact K — a Banach space

Nonwandering set
 $z \in \operatorname{NW}(g)$: every neighborhood $U \ni z$, $\forall t_0 \exists t > t_0$: $\varphi_t(U) \cap U \neq \emptyset$.

Peixoto compact 2-manifold
 $f \in C^1$ structurally stable $\iff$
• **(T1)** finitely many equilibria & periodic orbits, all hyperbolic
• **(T2)** no saddle–saddle connections, $W^s(p) \cap W^u(q) = \emptyset$
• **(T3)** $\operatorname{NW}(g) = \{\text{equilibria}\} \cup \{\text{periodic orbits}\}$
 M orientable $\Rightarrow$ structurally stable fields are open and dense in $C^1(M)$. Fails for $d \geq 3$.
ex. torus flow $x' = \omega_1 x$, $y' = \omega_2 y$: structurally unstable ($\omega_1/\omega_2 \in \mathbb{Q}$: all orbits periodic; irrational: all orbits dense)

Morse–Smale system
(D1) finitely many equilibria & periodic orbits, all hyperbolic; **(D2)** all $W^s \cap W^u$ transversal; **(D3)** NW = equilibria $\cup$ periodic orbits.
Thm: Morse–Smale flows on compact M are structurally stable (any dim) — but not generic for $d \geq 3$.

Generic property
holds on a **residual** set = countable intersection of open dense sets.
ex. hyperbolicity is generic in $\mathbb{R}^{d \times d}$

The road to chaos

§14, §15

Periodically perturbed Hamiltonian system
 $x' = f(x) + \varepsilon g(x, t)$, $f = (\partial_x H, -\partial_x H)^T$, g T -periodic; unperturbed homoclinic orbit $\gamma(t)$ to a saddle p . Automorize on $\mathbb{R}^2 \times S^1$, take the stroboscopic Poincaré map P_0^T — the saddle persists: $P_0^T = p + O(\varepsilon)$.

1. leading-order splitting of W^u , W^s

signed distance: $d(h) = \varepsilon \frac{M(h_0)}{2} + O(\varepsilon^2)$, $a \wedge b = a_1 b_2 - a_2 b_1$.

simple zero of $M \Rightarrow P_0^T = p$ has no intersection.
Duffing $x' = x - x^3 + \varepsilon(\alpha \cos t - x^2)$, $\gamma_1 = \pm(\sqrt{2} \operatorname{sech} t, -\sqrt{2} \operatorname{sech} t \tanh t)$: $M(h_0) = -\frac{1}{2} + \sqrt{2} \cos \operatorname{sech}(\frac{1}{2}) \sin \alpha$ — simple zeros for α large enough
1. transverse homoclinic point $q \in W^u(0) \cap W^s(0)$

Smale–Birkhoff homoclinotic theorem
diffeo with hyperbolic saddle + transverse homoclinic point $\Rightarrow$ some iterate g^n has an invariant set X conjugate to a **horseshoe**.
 λ forms a disk D transverse to W^s that iterates $g^i(D)$ accumulating C^0 on W^u — this folds a rectangle near q into a horseshoe. (Transverse homoclinic points occur only in maps / Poincaré maps, never in planar flows.)
1. stretch $\mu > 2$, contract $\lambda < \frac{1}{2}$, fold

Smale horseshoe $\geq$ full shift
 $X = \{x \colon g^i(x) \in Y = [0, 1]^2 \forall k \in \mathbb{Z}\}$; itinerary map $\varphi(x) = a_j$ with $g^j(x) \in H_{a_j}$ is a homeomorphism conjugating $g|_X$ to the shift σ on $\Sigma_2 = \{0, 1\}^{\mathbb{Z}}$.
generalized horseshoe: N strips + cone conditions $\Rightarrow$ shift on N symbols
• count orbits systematically

Singer
 $g \in C^1(I, \mathbb{R})$, $Sg < 0$, n critical points $\Rightarrow$ at most $n + 2$ attracting periodic orbits.
logistic $\operatorname{pr}(1 - x)$: $Sg = \frac{1-2x}{1-x^2} < 0$, one critical point $\Rightarrow \geq 3$

The bifurcation zoo

§8, §12, §13

Bifurcation loss of equivalence under parameter variation
Families $x' = f(x, p)$, $y' = g(y, q)$ are **fiber topologically equivalent** if a homeomorphism $\xi(p) = q$ matches parameters and each $f(\cdot, p) \sim g(\cdot, \xi(p))$ topologically. A **bifurcation diagram** stratifies parameter space by equivalence class; an **unfolding** contains the bifurcation persistently; the minimal number of parameters is the **codimension**.

